

Grade 7 Mathematics — Mock Exam — Paper C

HARD • NON-CALCULATOR • 75 MINUTES • 80 MARKS

NAME

DATE

SCORE / 80

- Answer **every** question in the space provided. Show your working — method marks are awarded even when the final answer is wrong.
- No calculator. Every number on this paper is designed to work without one.
- **75 minutes for 80 marks** — about a minute a mark. If a 3-mark question has taken you five minutes, leave it and come back to it.
- No formulas and no methods are given anywhere on this paper — not one. Remembering the right one is part of what is being tested.
- Give every fraction and every ratio in its simplest form, and convert to a common unit before you simplify.
- The mark for each question is shown in square brackets on the right.
- Nothing tells you which topic a question belongs to, and several need two topics at once. Work out what you are being asked before you start writing.

SECTION 1 – QUICK FIRE

1. Write down the answer to each of these.

(a) Write down the largest prime number that is less than 40. [1]

(b) $-3 \times -4 \times -2$ [1]

(c) $\sqrt{81} + \sqrt[3]{8}$ [1]

(d) Write 4^3 as a power of 2. [1]

(e) $0.06 \div 0.01$ [1]

2. Write down the answer to each of these.

(a) Round **0.0498** to 2 decimal places. [1]

(b) 0.04×0.05 [1]

(c) Write $\frac{7}{11}$ as a decimal. Use dot notation. [1]

(d) Write the ratio 750 g : 1.5 kg in its simplest form. [1]

(e) Factorise fully $12x^2 - 18x$. [1]

SECTION 2 – SHORT ANSWERS

3. Two numbers are 24 and 36.

(a) Find the highest common factor (HCF) of 24 and 36. [1]

(b) Find the lowest common multiple (LCM) of 24 and 36. [1]

4. Simplify $(2^3 \times 2^5) \div (2^2 \times 2^4)$. Give your answer as an ordinary number. [2]

5. Work out:

(a) $3.6 \div 0.01$

[1]

(b) 0.07×0.1

[1]

6. Write these decimals in descending order — largest first.

[2]

0.4 0.401 0.41 0.0409

7. Write these fractions in ascending order — smallest first. You must show your working.

[2]

$\frac{5}{6}$ $\frac{7}{9}$ $\frac{11}{12}$ $\frac{13}{18}$

8. Work out $6\frac{1}{6} - 2\frac{7}{8}$. Give your answer as a mixed number in its simplest form.

[2]

9. A rectangle is 3 cm longer than it is wide. Its width is w cm. Write an expression, in terms of w and in its simplest form, for the perimeter of the rectangle.

[2]

10. Expand and simplify $3(2x - 5) - 4(3 - x)$. [2]

SECTION 3 – MULTI-STEP

11. Work out:

(a) -7×-3 [1]

(b) $-7 - (-3)$ [1]

(c) $-60 \div -4$ [1]

12. Work out $0.918 \div 0.027$, showing your method. [3]

13. Work out $15 \times 2\frac{4}{5}$. [3]

14. Work out $9 \div \frac{3}{8}$. [3]

15. The perimeter P of a rectangle of length l and width w is given by $P = 2(l + w)$.

(a) Find P when $l = 9$ and $w = 6.5$. [1]

(b) Find w when $P = 46$ and $l = 13$. [2]

16. Expand and simplify $4(3x + 5) - 2(2x - 5)$, then factorise your answer fully. [3]

17. Solve $2(3x - 1) = 4(x + 3)$. Show every step. [3]

18. A television costs ₹1250. Its price is reduced by 18%. Work out the new price. [3]

SECTION 4 – PROBLEMS

19. ₹9450 is shared between three charities in the ratio 2 : 3 : 4.

(a) Work out how much each charity receives. [3]

(b) The largest share is then increased by 12%. Work out its new value. [1]

(c) Work out the new total amount shared between the three charities. [1]

20. The three angles of a triangle are x° , $(3x - 20)^\circ$ and $(x + 40)^\circ$.

(a) Form an equation in x and solve it. [3]

(b) Hence write down the size of the largest angle. [2]

21. A shop sells juice in three sizes.

0.75 litres for ₹57 1.25 litres for ₹90 2.5 litres for ₹185

(a) Work out the price per litre of each size. [3]

(b) Write down which size is the best value. [1]

(c) How much is saved by buying 5 litres in the best-value size rather than in the worst-value size? [1]

22. Machine A is serviced every 15 days. Machine B is serviced every 24 days. Both machines are serviced on 1 March.

(a) Work out how many days pass before both machines are next serviced on the same day. [3]

(b) How many times in the 365 days after 1 March are both machines serviced on the same day? [2]

23. A share is worth ₹2000. In week 1 its value rises by 30%. In week 2 its new value falls by 30%.

(a) Work out its value at the end of week 1. [1]

(b) Work out its value at the end of week 2. [2]

(c) Work out the overall percentage change over the two weeks. [2]

24. A rectangle has length $(3x + 2)$ cm and width 4 cm.

(a) Write an expression for its area. Expand your answer. [2]

(b) Write an expression for its perimeter, in its simplest form. [2]

(c) Factorise your expression for the perimeter fully. [1]

End of paper. Go back to the two questions you were least sure of. On a mock, the marks you find in the last five minutes are the cheapest ones on the paper.